

Missing Data Mechanism and Mean Imputation

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Income Dataset

Load the required libraries

```
library(MASS)
library(tidyverse)
library(ggplot2)

# set seed for reproducibility
set.seed(45)
```

Explore the data

```
# Load data
income <- readRDS("income.rds")

# first 5 rows
head(income, 5)
```

```
##   age gender income
## 1  33 Female   2111
## 2  59 Female   2214
## 3  39 Female   2478
## 4  64 Female   2598
## 5  67   Male   2167
```

```
# columns
names(income)
```

```
## [1] "age"      "gender"   "income"
```

```
# five number summary
summary(income)
```

```
##           age           gender           income
##  Min.   :18.00   Length   :500   Min.     :1613
##  1st Qu.:31.00   N.unique  : 2   1st Qu. :2165
##  Median :43.00   N.blank  : 0   Median  :2307
```

```
## Mean      :43.74   Min.nchar: 4   Mean      :2306
## 3rd Qu.   :56.00   Max.nchar: 6   3rd Qu.   :2461
## Max.      :70.00                               Max.      :2951
```

```
# Dataset consists of 3 variable i.e gender, age and income
# Average income and age is 2306 dollar and 43 years respectively
```

Function for computing the probability of a value being missing for each person

using logistic regression model. This function takes a person's age, gender, and income, then weights each one by ψ values, and returns a number between 0 and 1 representing that person's chance of having a missing income value. Setting different ψ values is exactly how the practical simulate:

- NDD (all ψ = 0, constant 0.5 for everyone)
- SDD (nonzero ψ_1/ψ_2 , zero ψ_3 , depends only on observed age/gender)
- UDD (nonzero ψ_3 , depends on income itself).

```
prob_mis <- function(psi0, psi1, psi2, psi3, age, gender, income) {
  exp(psi0 + psi1 * age + psi2 * (gender == "Male") + psi3 * income) /
  (1 + exp(psi0 + psi1 * age + psi2 * (gender == "Male") + psi3 * income))
}
```

Not Data Dependent (NDD)

```
income <-
  income |>
  mutate(misprob_ndd = prob_mis(psi0 = 0,
                                psi1 = 0,
                                psi2 = 0,
                                psi3 = 0,
                                age = age,
                                gender = gender,
                                income = income))
```

```
income <-
  income |>
  mutate(R_ndd = 1 - rbinom(n(), 1, misprob_ndd))
```

```
income <-
  income |>
  mutate(income_ndd = ifelse(R_ndd == 0, NA, income))
```

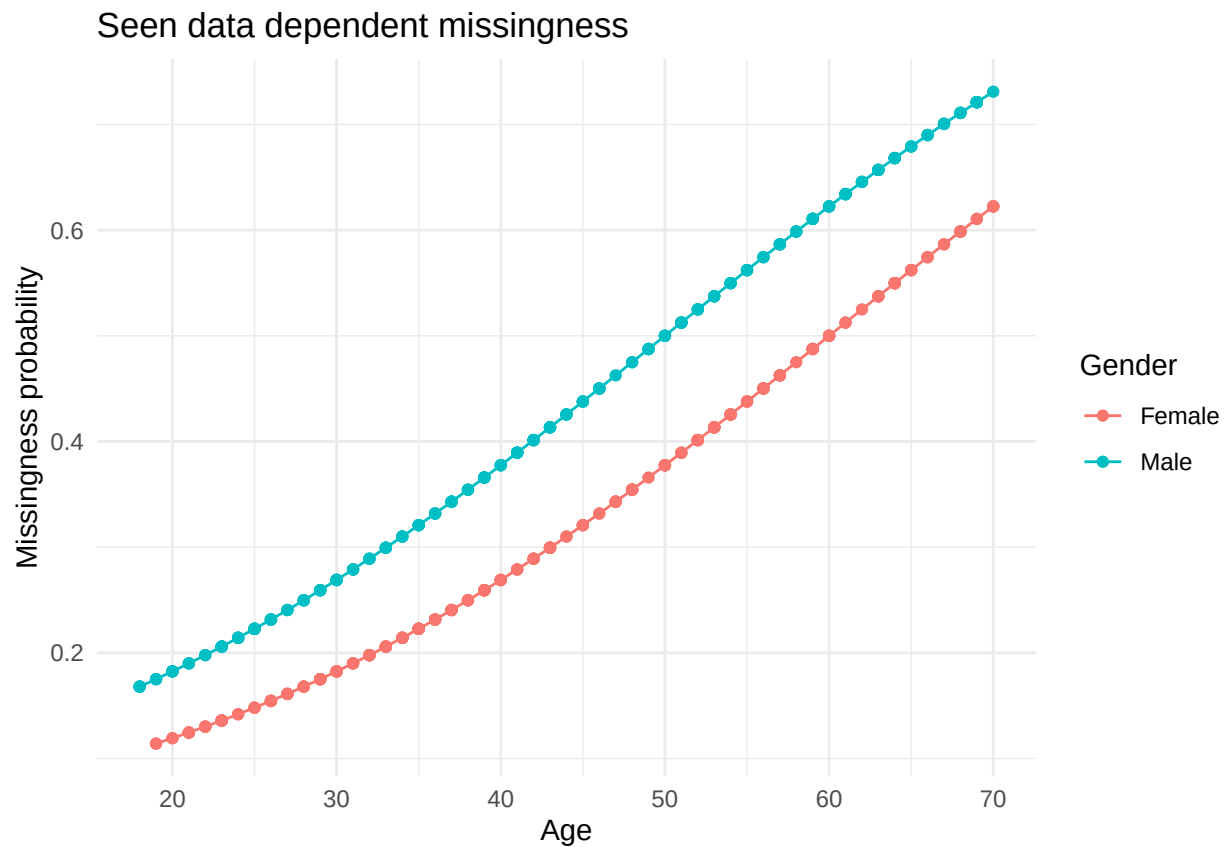
```
table(is.na(income$income_ndd))
```

```
##
## FALSE TRUE
##   263   237
```

Seen Data Dependent (SDD)

```
income <-  
  income |>  
  mutate(misprob_sdd = prob_mis(  
    psi0 = -3,  
    psi1 = 0.05,  
    psi2 = 0.5,  
    psi3 = 0,  
    age = age,  
    gender = gender,  
    income = income))
```

```
income |>  
  ggplot(aes(x = age, y = misprob_sdd, colour = gender)) +  
  geom_line() +  
  geom_point() +  
  theme_minimal() +  
  labs(  
    x = "Age",  
    y = "Missingness probability",  
    colour = "Gender",  
    title = "Seen data dependent missingness"  
  )
```



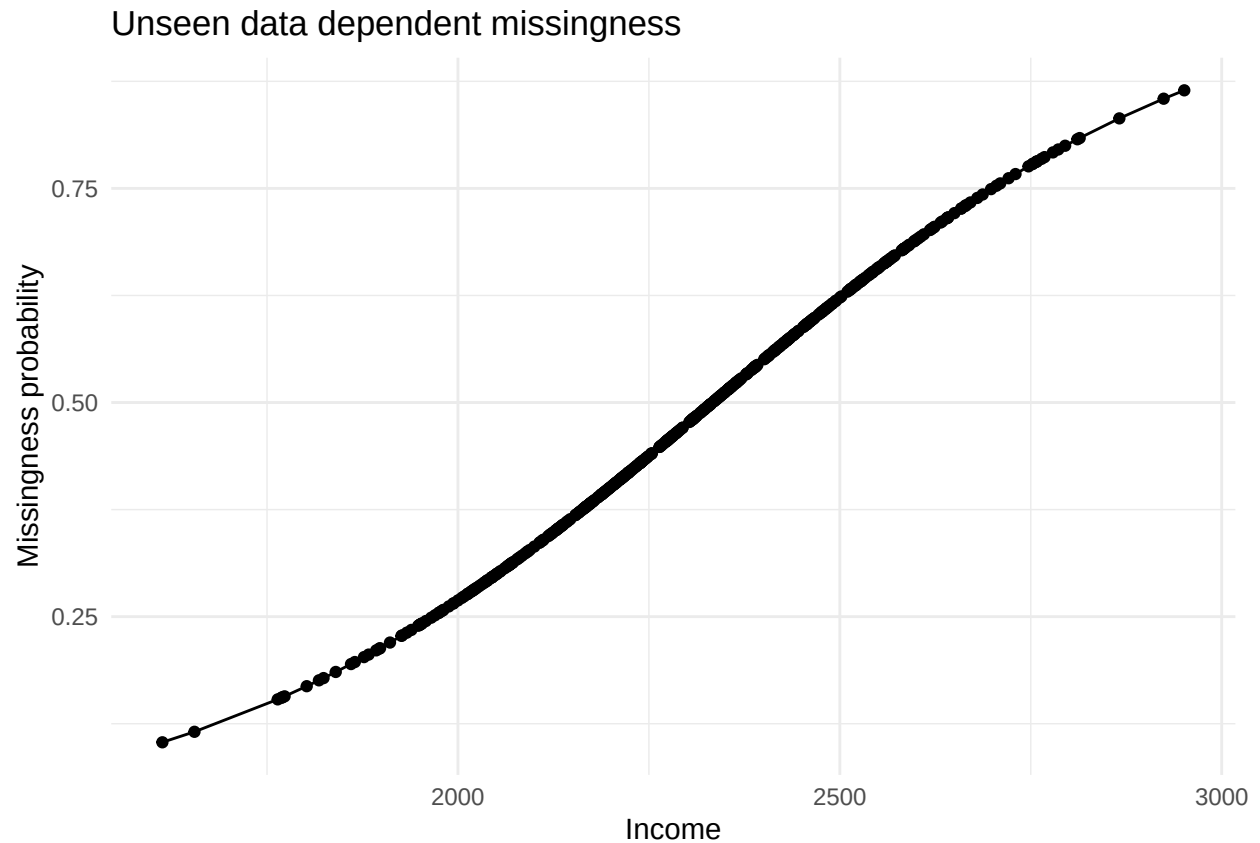
```
# missingness probability rises steadily as age increases, with males consistently
# showing a higher probability than females at every age.
# This is as expect given the parameters we set (psi1 = 0.05 for age, psi2 = 0.5 for male).
# It confirms the SDD mechanism is working correctly because missingness depends
# on observed variables (age and gender).
# An 18-year-old female has roughly a 10% chance of missing income, while a
# 70-year-old male has over 70%.
```

```
income <-
  income |>
  mutate(
    R_sdd = 1 - rbinom(n(), 1, misprob_sdd),
    income_sdd = ifelse(R_sdd == 0, NA, income)
  )
```

Unseen Data Dependent (UDD)

```
income <-
  income |>
  mutate(misprob_udd = prob_mis(
    psi0 = -7,
    psi1 = 0,
    psi2 = 0,
    psi3 = 0.003,
    age = age,
    gender = gender,
    income = income
  ))
```

```
income |>
  ggplot(aes(x = income, y = misprob_udd)) +
  geom_line() +
  geom_point() +
  theme_minimal() +
  labs(
    x = "Income",
    y = "Missingness probability",
    title = "Unseen data dependent missingness"
  )
```



*# Missingness probability climbs steeply as income itself increases, from about 10%
at the lowest incomes to nearly 90% at the highest.
This is the defining feature of UDD where the probability of a value being missing
depends on the value itself.
High earners are much more likely to have their income hidden.*

```
income <-  
  income |>  
  mutate(  
    R_udd = 1 - rbinom(n(), 1, misprob_udd),  
    income_udd = ifelse(R_udd == 0, NA, income)  
  )
```

Visualize missing data

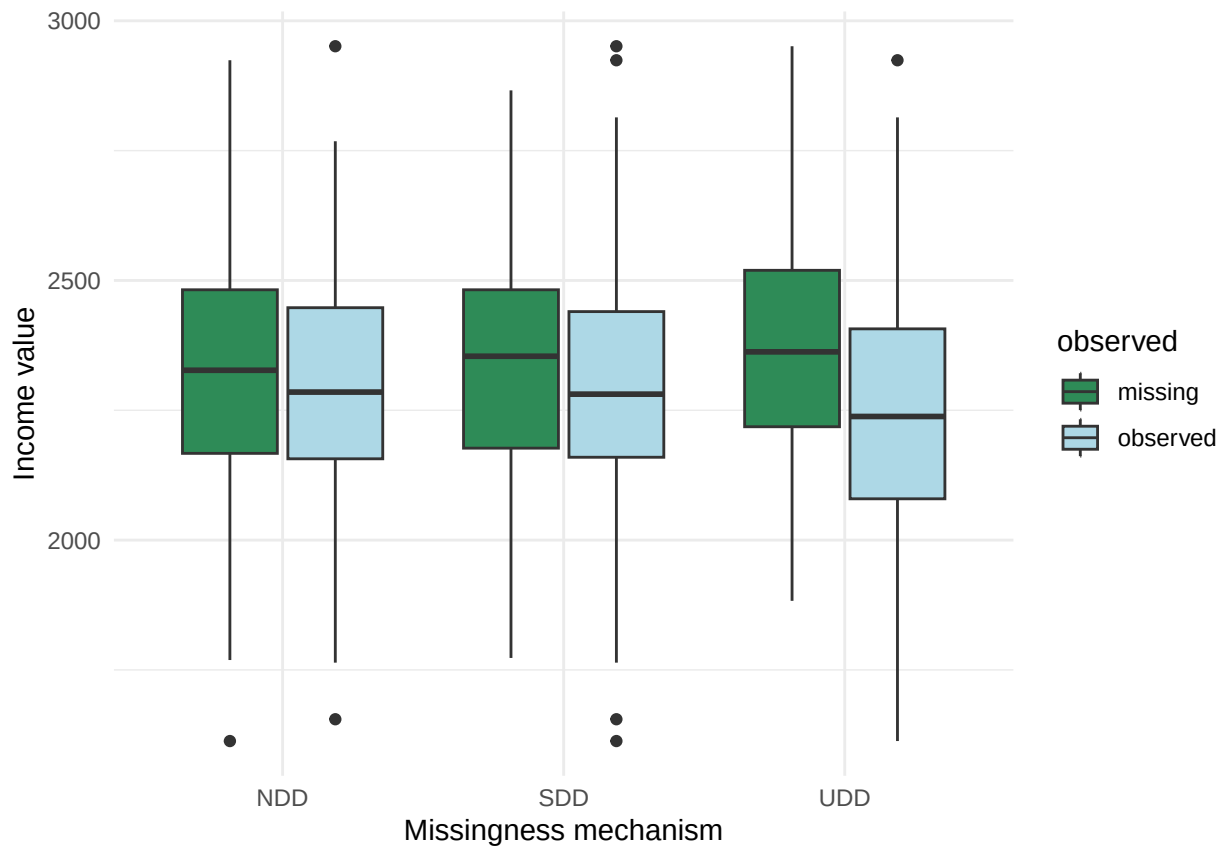
Visualization can help in understand the missing data mechanism by providing insight into the amount and pattern of missingness.

```
income |>  
  pivot_longer(  
    cols = c(R_ddd, R_sdd, R_udd),  
    names_to = "mechanism",  
    values_to = "observed",  
    names_prefix = "R_"
```

```

) |>
mutate(observed = factor(ifelse(observed == 0, "missing", "observed"))) |>
ggplot(aes(x = str_to_upper(mechanism), y = income, fill = observed)) +
geom_boxplot() +
theme_minimal() +
scale_fill_manual(values = c("seagreen", "lightblue")) +
labs(
  x = "Missingness mechanism",
  y = "Income value",
  colour = ""
)

```



*# These boxplots display the distributions of \$Y_1\$ for the observed and the
 # missing group under the three different missing data mechanisms. We see that
 # under NDD, the distributions are approximately similar, while they become
 # progressively more distinct under SDD and UDD.*

Mean Imputation

Why mean imputation, in general, is a bad idea

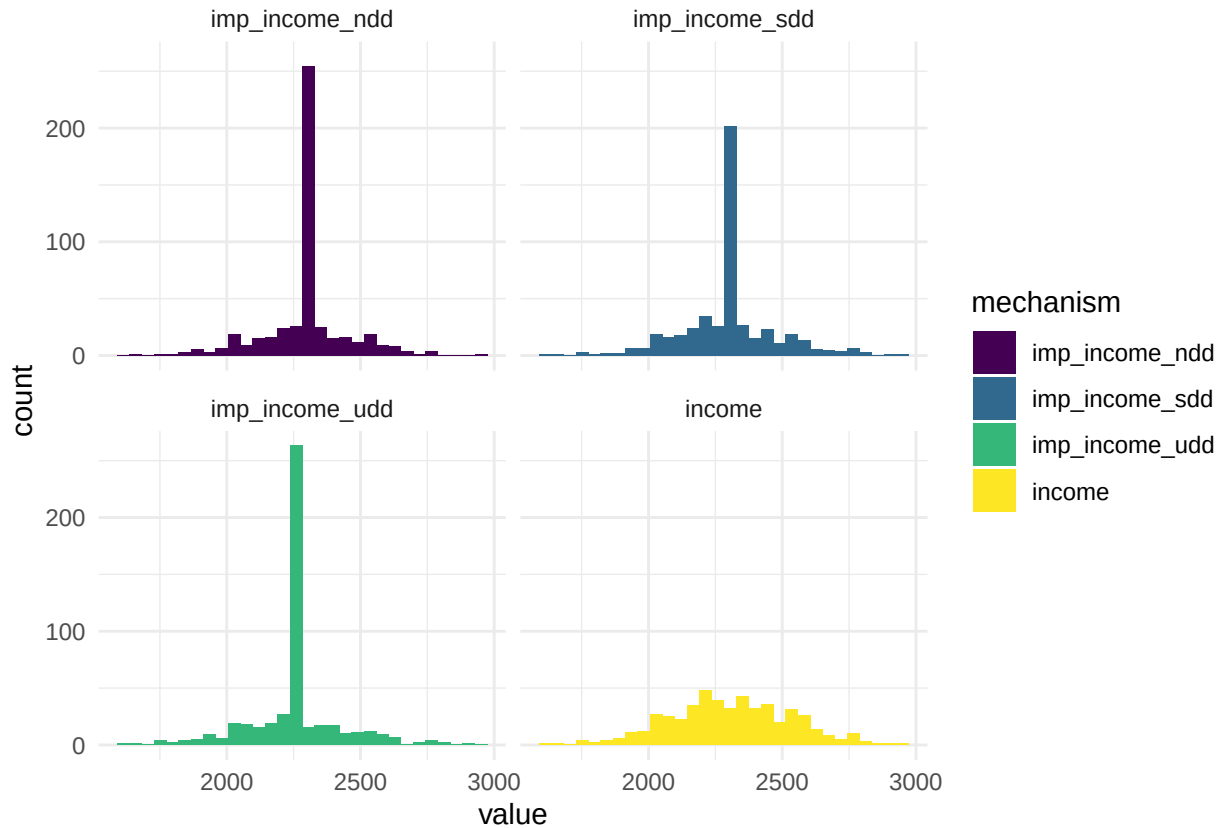
with across() we can apply a mutate over several columns

```
income <-
  income |>
  mutate(across(
    .cols = c(income_ndd, income_sdd, income_udd),
    .fns = function(x) replace_na(x, mean(x, na.rm = TRUE)),
    .names = "imp_{.col}"
  ))
```

```
income |>
  pivot_longer(
    cols = c(income, starts_with("imp_")),
    names_to = "mechanism",
    values_to = "value"
  ) |>
  group_by(mechanism) |>
  summarize(mean = mean(value), sd = sd(value))
```

```
## # A tibble: 4 x 3
##   mechanism      mean    sd
##   <chr>         <dbl> <dbl>
## 1 imp_income_ndd 2291.  155.
## 2 imp_income_sdd 2297.  180.
## 3 imp_income_udd 2250.  165.
## 4 income        2306.  224.
```

```
income |>
  pivot_longer(
    cols = c(income, starts_with("imp_")),
    names_to = "mechanism",
    values_to = "value"
  ) |>
  ggplot(aes(x = value, fill = mechanism)) +
  geom_histogram() +
  facet_wrap(vars(mechanism)) +
  theme_minimal() +
  scale_fill_viridis_d()
```



```
# The mean imputation creates an enormous "peak" in the middle that is not there
# in the real data.
# mean imputation shrinks the standard deviation and distorts the shape of the distribution,
# misrepresenting the real spread of incomes.
# this problem exists for all three mechanisms, which is why mean imputation is
# described as "in general, a bad idea" regardless of what caused the missingness.
```

```
head(income, 10)
```

```
##      age gender income misprob_ndd R_ndd income_ndd misprob_sdd R_sdd income_sdd
## 1    33 Female  2111         0.5     0         NA    0.2058704     1      2111
## 2    59 Female  2214         0.5     1      2214    0.4875026     1      2214
## 3    39 Female  2478         0.5     1      2478    0.2592251     1      2478
## 4    64 Female  2598         0.5     1      2598    0.5498340     0         NA
## 5    67  Male  2167         0.5     1      2167    0.7005671     0         NA
## 6    20  Male  2121         0.5     1      2121    0.1824255     1      2121
## 7    45  Male  2136         0.5     1      2136    0.4378235     1      2136
## 8    64  Male  2034         0.5     0         NA    0.6681878     1      2034
## 9    47 Female  2359         0.5     1      2359    0.3429895     1      2359
## 10   42  Male  2278         0.5     1      2278    0.4013123     1      2278
##      misprob_udd R_udd income_udd imp_income_ndd imp_income_sdd imp_income_udd
## 1    0.3391689     0         NA      2291.049      2111.000      2250.21
## 2    0.4114438     0         NA      2214.000      2214.000      2250.21
## 3    0.6068284     1      2478      2478.000      2478.000      2478.00
## 4    0.6886896     1      2598      2598.000      2296.545      2598.00
```

## 5	0.3777757	1	2167	2167.000	2296.545	2167.00
## 6	0.3459250	1	2121	2121.000	2121.000	2121.00
## 7	0.3561761	0	NA	2136.000	2136.000	2250.21
## 8	0.2894617	1	2034	2291.049	2034.000	2034.00
## 9	0.5192405	1	2359	2359.000	2359.000	2359.00
## 10	0.4585950	0	NA	2278.000	2278.000	2250.21